Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4.	*(a)*		**Requirements** • Direction of polarisation in common • Comparable amplitude • Same frequency / wavelength • Coherent • Diffraction **Possible explanation material** • Coherent sources means constant [**or** constant or zero] phase difference between sources • Sketch-graphs showing meaning of constant phase difference • Rapidly shifting interference pattern if phase difference varies • No cancellation if polarisation directions or amplitudes different • Diffraction needed for overlap **Possible examples of compliance and non-compliance** • Two *slits* illuminated by single laser would fulfil requirements • Slits illuminated by ordinary lamp or separate lamps wouldn't • Contrived cases e.g. sources polarised at 90^{o} to each other; polaroids crossed or one source (e.g. slit) covered by dark glass acceptable	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			5-6 marks Comprehensive account of all areas i.e. requirements, explanations and examples given. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** Comprehensive account of 2 out of the 3 areas i.e. requirements, explanations and examples given **or** limited attempt of all 3 areas. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** Comprehensive account of 1 out of the 3 areas i.e. requirements, conditions and examples given **or** limited attempt of 2 areas. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** No attempt made or no response worthy of credit.						

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
	(b)	(i)	Mean of measurements i.e. x (= 5Δy) = 6.6[2] [mm] **or** by impl (1) Mean of Δy = 1.3[2] [mm] (1) Absolute uncertainty in x = 0.3 [mm] **or** in Δy = 0.06 [mm] or by implication (1) % uncertainty = 5 % Accept 4.5 % **ecf** on Δy and unc in Δy and on x and unc in x (1)		4		4	3	4
		(ii)	Correct substitutions into $\lambda = \frac{a\,\Delta y}{D}$, with **ecf** on Δy and allowing slips in powers of 10 (1) λ = 520 n[m] or 528 n[m] with **ecf** on Δy (1) % uncertainty = 10 % Accept 9.5 % **ecf** on uncertainty in Δy (1)		3		3	3	3
	(c)	(i)	$\lambda = \frac{1500\text{ nm}\sin 45°}{2}$ substitution in re-arranged eq. allow slips of powers of 10 (1) λ = 530 n[m] Accept 530.3 n[m] (1)		2		2	2	
		(ii)	Bright areas further apart in case of grating (1) Because slits closer together in grating [than double slits] (1) not freestanding Bright areas sharper **or** smaller **or** brighter for grating or equivalent (1) Because more slits in grating **or** more destructive interference **or** (for brighter fringes) more constructive interference (1) not freestanding	4			4		
			Question 4 total	**10**	**9**	**0**	**19**	**8**	**7**